

Supplementary Information: IRSpectra-Bench — protocol, scale tables, cases, and reproducibility

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Companion to the ICLR 2027 submission

Generation Recall, Not Verification, Binds Molecular Structure Elucidation
from Literature Peak Lists

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This SI holds volume that does not belong on the 9-page ICLR main text. Every accuracy number is copied from a committed lock (docs/HEADLINE_N500_2026-09-20.md, spectro-agent PR #41 48fcd30; docs/NIGHT_POOL_2026-09-20.md). No CIs are invented for $n=500$. The paper headline is $n=500$ (227/500 top-1; 249/500 recall). The diagnosis wall is the $n=500$ forward-verify decomposition 204/45/251 (spectro-agent data/fverify_n500/WALL_n500.md). Generation 227/249/251 is top-1 / recall / never-proposed, not the wall. Do not quote 243/519. Roster: frozen qid lists in the code release. Figure and listing numbers below refer to the main PDF (Fig. 1 = protocol + $n=500$ fverify; Listing 1 = IRexp payload; Fig. fig-wall = $n=500$ fverify wall). They are not re-numbered here.

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1 How to read this SI (figure / listing map)

The night main-text plate and listing are the source of truth. This SI does not fight that numbering.

Equations in the main text are Eqs. (1)–(7): factorisation, InChIKey-14 indicator, top-1, recall@ k , prec.|recall, ¹³C chamfer, pooled estimator. They are restated in §4 with the same meaning. No new metric is defined.

2 Visual plates (reprints; not a second Figure 1)

The files below are the same as the main PDF. They are reprinted so this SI is readable alone. Main-text Figure 1 (fig1_lead_overview) remains the only lead diagram. Panel letters (a)/(b)/(c) from an earlier draft do not apply: the Figma Bro plate is one figure (protocol strip, factorisation, wall).

3 Blind protocol and pre-registration

3.1 Solver-facing contract

Each IRSpectra-Bench item is a four-field JSON object. The solver sees only

Table 1: Main-text floats vs this SI. Fig. 1 is the Figma Bro lead plate.

main label	object	SI role
Fig. fig1	Fig. 1 (fig1_lead_overview)	Lead: input $\rightarrow$ generate $\rightarrow$ select, factorisation, and $n=500$ fverify (249/500 recall $\times$ 204/249 verified pool = 40.8% verified; wall 204/45/251).
Fig. fig-chemspace	Fig. 2	Literature-cohort chemistry.
Listing lst:payload	Listing 1 (listing1_irexp_record)	IRexp JSON+molecule (NTJIYHYWVZY-BEX; C ₁₂ H ₁₆ N ₂ O ₂). SMILES/InChIKey are resource fields, not solver payload.
Fig. fig-wall	appendix (fig_wall_diagnostic)	$n=500$ forward-verify wall 204/45/251.
Fig. fig-robustness	Fig. 4	Formula-only / recency / four-vendor.
Fig. fig-mechanism	Fig. 5	Locked v3-R25 verify-save. Reused in §9.
Fig. fig3-method	Fig. 6	Inference ladder on the 60-compound arm.
App. cases	worked modes	Same integers as §9.
SI Figs. S1-S3	reprints	Same three files as main Fig. 1 / Listing 1 / $n=500$ fverify wall (§2). Not a second lead figure.

{qid, formula, ir_bands_cm-1, h_nmr, c_nmr}.

No name, SMILES, InChIKey, scaffold, atom-type hint, retrieved spectrum, licence stamp, or PMC id enters the prompt (main-text Listing 1, adapted from one IRexp record). Gold constitution is scored offline by the InChIKey-14 indicator (§4).

Difficulty is declared a priori by RDKit ring analysis: simple iff ≤ 2 rings, no fused/spiro/bridge-head system, ≤ 22 heavy atoms; else complex. The $n=500$ headline is near-50/50 (248/252); the eligible IRexp corpus is 17.5% simple / 82.5% complex. Reweighting lives on a validate-clean subset (§5) and is not an $n=500$ figure.

3.2 Pre-registration pointers

These files live in the companion spectro-agent tree, not this postcard manuscript repo. Pointers are recorded so a reviewer can ask for the committed note rather than a reconstructed story.

- $n=500$ roster: frozen qid lists in the code release (do not reconstruct a composition story in prose). Sampler seeds for early draws were not captured (gap #4 in docs/SUBMISSION.md); the released roster is fixed and reusable but not exactly re-drawable.
- expand-500 campaign: docs/EXPANSION_PREREGISTRATION_500.md (seed 2026500; 115/115 simple/complex). 0 collisions with any prior round. Six ¹³C-overread flags known before any solve: R26, R31, R102, R105, R107, R138.

3.3 Stopping / pooling rules

- The $n=500$ roster is the pre-specified frozen list (docs/headline500_expand200_qids.json plus the committed deposits), not an outcome-selected subset.
- score_pooled.py –expand-500 refuses a missing predictions2.jsonl or a withheld key.
- answers2.jsonl is not in either working tree (git ls-files empty for the withheld expansion keys).
- Keys were restored only under /tmp/blind/_key/ for the 2026-09-20 score pass, then deleted.
- validate_benchmark.py was not re-run on the night pass (would rewrite committed clean_qids.json). The existing 224-qid snapshot is used.



Figure S1: Reprint of main-text Figure 1 (fig1_lead_overview). Generate 249/500 (49.8%) recall in top 3; select 204/249 (81.9%) verified | pool (49.8% × 81.9% = 40.8% verified). Wall: 204 verified / 45 misranked / 251 never proposed. Propose is the wall, not verify.

- collect_round.py was not run; 230 raw/single_R*.json files were left untouched.
- forward_verify_main.py score did run on expand-500 (§8): 103/230 verify vs 129/230 self. Official prep was not re-run.
- -ci / bootstrap was not run on expand-500. Do not invent CIs.

3.4 What the solver may do

RDKit formula / parse check only. No web search. No retrieved spectrum. No thinking tier on the core protocol. A thinking-tier consumer arm is labelled as such in the score notes and is not interchangeable with the no-thinking protocol.

A deposit is JSONL of {qid, candidates: [SMILES,...]} under a stated budget. Headline budget: ≤ 3 ranked SMILES per qid. Self-ranking vs forward-verified re-ranking must be labelled; they are not interchangeable.

4 Scoring contract (equations used in the main text)

These are the same identities as main-text Eqs. (1)–(7). They are restated so the SI is readable alone. No new metric is introduced.

A candidate $\hat{y}$ matches gold y^* at constitution iff

$$s(\hat{y}, y^*) = \mathbf{1}[\text{IK}_{14}(\hat{y}) = \text{IK}_{14}(y^*)]. \quad (1)$$

IK₁₄ is the first 14 characters of the RDKit InChIKey (connectivity layer). Full stereo-sensitive InChIKey is reported only on validate-clean $n=295$ (§5).

Listing 1. Example IRexp JSON record (commercial DoR).

```
{
  "id": "NTJIYHYWVZYBEX-UHFFFAOYSA-N",
  "ir_bands_cm-1": [3337.0, 2969.0, 1633.0, 1556.0,
    1505.0, 1240.0, 1181.0, 1071.0, 826.0],
  "ir_source": "experimental",
  "source_doi": "PMC:13029360",
  "pmcid": "PMC13029360",
  "h_nmr": "8.07 (brs, 1H, NH), 7.25 (d, J = 7.3 Hz, 2H, Ar), 6.78 (d,
    J = 7.3 Hz, 2H, Ar), 6.25 (d, J = 8.1 Hz, 1H, NH), 4.10
    (hex, J = 8.1 Hz, 1H, CH), 3.66 (s, 3H, OCH3), 2.23-2.09
    (m, 2H, CH2), 1.90-1.71 (m, 2H, CH2), 1.64-1.51 (m, 2H,
    CH2)",
  "c_nmr": "154.8, 154.4, 134.0, 119.9, 114.3, 55.5, 45.0, 31.8, 14.8",
  "smiles": "C0c1ccc(NC(=O)NC2CCC2)cc1",
  "inchikey": "NTJIYHYWVZYBEX-UHFFFAOYSA-N",
  "has_structure": true,
  "license": "CC-BY",
  "license_pool": "commercial"
}
```

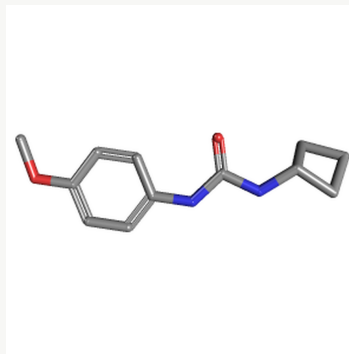


Figure S2: Reprint of main-text Listing 1 (listing1_irexp_record). IRexp JSON+molecule ($C_{12}H_{16}N_2O_2$; NTJIYHYWVZYBEX-UHFFFAOYSA-N). SMILES / InChIKey are resource fields, not solver payload. Not a Data Descriptor dump. Locked v3-R25 is the mechanism case (Figure S4), not this listing.

With ranked pool $\mathcal{C}_i = (c_{i1}, \dots, c_{ik})$, $k \leq 3$,

$$\text{top-1} = \frac{1}{n} \sum_i s(c_{i1}, y_i^*), \quad (2)$$

$$\text{recall}@k = \frac{1}{n} \sum_i \mathbf{1}[\exists j \leq k : s(c_{ij}, y_i^*) = 1], \quad (3)$$

$$\text{prec.}|\text{recall} = \frac{\sum_i s(c_{i1}, y_i^*)}{\sum_i \mathbf{1}[y_i^* \in \mathcal{C}_i]}. \quad (4)$$

These are the three reportable numbers. They factorise as

$$\text{top-1} = (\text{recall}@k) \times (\text{prec.}|\text{recall}) \quad (5)$$

when the same pool is used for both stages — the rates still have different denominators and must not be differenced. Self-ranking prec.|recall on $n=500$ is 227/249 (91.2%). That is not forward-verify precision.

Forward-verify re-ranks by symmetric chamfer on ^{13}C peak sets $P, Q \subset \mathbb{R}$

$$d(P, Q) = \frac{1}{|P|} \sum_{p \in P} \min_{q \in Q} |p - q| + \frac{1}{|Q|} \sum_{q \in Q} \min_{p \in P} |q - p|, \quad (6)$$

where $\hat{P}(c)$ is predicted blind to the observed list P_{obs} . Instrumented forward-verify diagnostic on a protocol slice (58/65; §8). The $n=500$ forward-verify wall is 204/45/251 (§8; data/fverify_n500/WALL_n500.md). expand-500 official arm score is also in §8 (103/230 verify vs 129/230 self; no CIs; not a second wall).

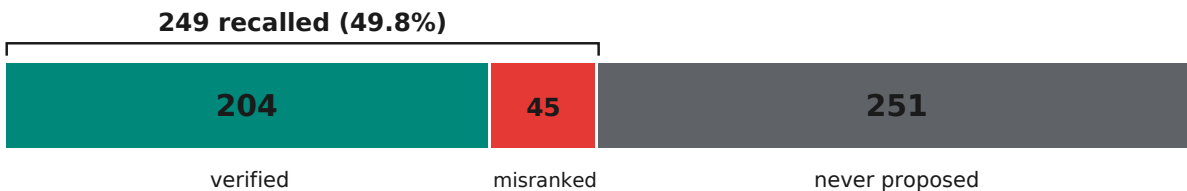


Figure S3: Reprint of the $n=500$ forward-verify wall (fig_wall_diagnostic): 204 verified / 45 mis-ranked / 251 never proposed. Source: spectro-agent data/fverify_n500/WALL_n500.md. Propose is the wall, not verify. Generation remains 227/500 top-1 and 249/500 recall.

A pool estimator on an index set $\mathcal{S}$ is

$$\hat{p}_{\mathcal{S}} = \frac{1}{|\mathcal{S}|} \sum_{i \in \mathcal{S}} s(c_{i1}, y_i^*). \quad (7)$$

The ICLR headline uses $\mathcal{S}$ of size 500 (frozen qid list). Do not quote the $n=519$ score_pooled.py print (243/519) as the headline.

Confidence intervals on earlier validate-clean subsets are bootstrap 95% (score_main.boot, seed 0, 2000 resamples). Paired model differences use McNemar with Holm correction. No CIs are reported for $n=500$ or expand-500.

5 Validate-clean extras (not the headline)

Source: docs/POOLED_HEADLINE_2026-09-16.md. These rows are not the $n=500$ headline. Self-ranking precision | recall on $n=500$ is 227/249 (91.2%) — not forward-verify precision.

¹³C-overread flags (R12, R22, R25, R82, R91) were marked before subset totals (scripts/validate_benchmark.py data/benchmark_expand/clean_qids.json).

Table 2: Validate-clean extras. Same scorer; not the headline.

metric		overall	simple	complex
scaffold-level (best Tanimoto ≥ 0.45)		64%	80%	48%
mean best Tanimoto		0.66	0.79	0.53
full InChIKey (stereo) top-1	93/295 (31.5%)	[26–37]	—	—
full InChIKey recall	108/295 (36.6%)	[31–42]	—	—
corpus-reweighted top-1	26.5%	[21–32]	—	—
corpus-reweighted recall	31.3%	[25–37]	—	—

Table 3: Size bins on validate-clean $n=295$.

heavy atoms	n	top-1	recall
≤ 15	53	36/53 (67.9%)	39/53 (73.6%)
16–25	152	65/152 (42.8%)	75/152 (49.3%)
> 25	90	15/90 (16.7%)	16/90 (17.8%)

On an instrumented slice, of 137 analysable top-1 misses, 76.6% are constitutional isomers of the true structure; only 22.6% share the true Murcko scaffold. Named-ring annotation on that slice is rare and does not drive the result.

6 expand-500 generation (NIGHT_POOL)

Source: spectro-agent data/benchmark_expand_500/NIGHT_POOL_2026-09-20.md on cursor/expand-bench-500-b78b (copied to docs/NIGHT_POOL_2026-09-20.md). Key restored only for score2, then deleted. answers2.jsonl is not in the tree.

Table 4: expand-500 generation (score2, InChIKey-14). No CIs.

set	<i>n</i>	top-1	recall@3
all deposited Opus	230	129/230 (56.1%)	138/230 (60.0%)
simple	115	82/115 (71%)	87/115 (76%)
complex	115	47/115 (41%)	51/115 (44%)
validate-clean	224	127/224 (56.7%)	135/224 (60.3%)
simple	110	81/110 (74%)	85/110 (77%)
complex	114	46/114 (40%)	50/114 (44%)

Mean best Tanimoto (score2 footer, all 230): simple 0.863; complex 0.645. Flagged qids (in 230, out of 224): R26, R31, R102, R105, R107, R138. Deposits: 230/230; predictions2.jsonl 230 lines, 690 candidates. Unique (formula, IR, ¹³C) vs irexp_resolved.jsonl.gz: 230/230; 0 prior-round InChIKey-14 collisions. Protocol: thinking-tier consumer arm — not interchangeable with the no-thinking core protocol.

Verbatim score2 footer (do not polish the integers):

overall recovered (top-3): 138/230 (60%) top-1 exact: 129/230 (56%)
 simple : recovered 87/115 (76%) top1 82/115 (71%) meanBestTani 0.863
 complex : recovered 51/115 (44%) top1 47/115 (41%) meanBestTani 0.645

7 Headline pool (generation only)

The estimator is Eq. 7. The paper headline is $n=500$. Source: docs/HEADLINE_N500_2026-09-20.md (spectro-agent 48fcd30). Roster: frozen qid lists in the code release (docs/headline500_expand200_qids.jsonl)

Table 5: Headline generation. No CIs. No fverify. No composition column.

set	<i>n</i>	top-1	recall@3
paper headline	500	227/500 (45.4%)	249/500 (49.8%)

Strata on the $n=500$ headline (same InChIKey-14 rule): simple 161/248 (65%) top-1 / 172/248 (69%) recall; complex 66/252 (26%) / 77/252 (31%). Self-rank precision | recall: 227/249 (91.2%). Do not quote score_pooled.py -expand -expand-500 ($n=519$, 243/519). The $n=500$ forward-verify wall is 204/45/251 (§8), not these generation rows. Do not attach CIs.

8 Forward-verify coverage (honest gaps)

8.1 $n=500$ unified fverify wall

Source: spectro-agent data/fverify_n500/WALL_n500.md (PR #68; scripts/fverify_n500_wall.py). Classification matches the sidecar: verified = recalled and chamfer-ranked first; misranked = recalled, distractor first; never-proposed = true structure not in the candidate set. The paper wall is the $n=500$ row only. Arm rows below are not a generation-composition story and are not the headline.

Table 6: $n=500$ forward-verify wall. Copied from data/fverify_n500/WALL_n500.md. No CIs. Arm rows are not the headline composition.

arm	n	verified	misranked	never-proposed
locked protocol slice	194	58	7	129
Opus expansion	106	56	12	38
expand-500 cut	200	90	26	84
$n=500$ wall	500	204	45	251

Recalled = $204 + 45 = 249/500$, matching committed generation recall. Main-text Figure fig-wall reprints this 204/45/251 wall. Figure 1 is the same $n=500$ fverify plate ($249/500$ recall $\times$ $204/249$ verified | pool = 40.8% verified; wall 204/45/251).

8.2 Instrumented protocol slice

Source: data/diagnosis.json / scripts/forward_verify_all.py. This is a protocol-slice diagnostic, not a second $n=500$ wall.

Table 7: Instrumented forward-verify diagnostic on a protocol slice. Not the $n=500$ wall (204/45/251).

	60-compound arm	protocol slice
generation recall	19/60 (32%)	65 (34%)
top-1, solver self-ranking	14/60 (23%)	55 (28%)
top-1, forward-verified re-ranking	16/60 (27%)	58 (30%)
precision recall — fverify	16/19 (84%)	58/65 (89%)
multi-candidate only — fverify	10/13 (77%)	30/37 (81%)
derangement floor on $n=37$	—	54.0%; fverify +27.1 points
McNemar vs self-rank	—	$p=0.55$ (7 gained, 4 lost)

8.3 Other expansion fverify — not a second wall

No additional expansion forward-verify bundle is claimed as a wall. Self-ranking 227/249 on $n=500$ is generation, not a substitute for 204/45/251.

8.4 expand-500 — official score (not a wall)

Source: spectro-agent data/fverify_expand_500/STATUS.md, results.txt, and NIGHT_POOL_2026-09-20.md (cursor/expand-bench-500-b78b). Blind ^{13}C campaign over 681 unique candidate SMILES

(f1-f41 all present). Official forward_verify_main.py score() ran 2026-09-20 (is_true filled under /tmp only; committed candidates.jsonl still omits it; official prep was not re-run). No CIs. Verify does not beat self-rank. Headline generation is $n=500$ (227/500, 249/500). This arm score is not a second wall and does not replace data/fverify_n500/WALL_n500.md.

Table 8: expand-500 official chamfer score. Copied from NIGHT_POOL / fverify STATUS. Arm score, not the $n=500$ wall. Not the headline.

set	n	recall	self top-1	verify top-1	verify recall
all deposited	230	138/230 (60.0%)	129/230 (56.1%)	103/230 (44.8%)	103/138 (74.6%)
validate-clean	224	135/224 (60.3%)	127/224 (56.7%)	103/224 (46.0%)	103/135 (76.3%)

Coverage: 41/41 batches; 681/681 unique SMILES; 230/230 qids fully covered. Multi-candidate only: all-230 self 128/137, verify 102/137; clean-224 self 126/134, verify 102/134. forward_verify_all.py was not run (no expand-500 diagnosis.json).

HOSE-style lookup on the protocol-slice candidate sets ties self-ranking; a modest GNN reaches 59/65 (91%) vs the LLM verifier’s 58/65 (89%) — directional, not a new headline. Deranging observed ^{13}C pairings collapses precision ($p=0.001$).

9 Failure and success case panels

The three locked-slice modes in the main appendix, plus extra panels that use only already-reported numbers. IDs below are locked-round labels (expansion qids reuse R22/R25). No SMILES are invented for R06/R26; only v3-R25 molecules are drawn (Figure S4).

Table 9: Three-mode case grid (disclosed facts only). Same integers as main-text Table tab:cases.

mode	id / formula	disclosed facts
Proposal miss	main-R06 / $\text{C}_{29}\text{H}_{30}\text{FNO}_2$	True acridinedione never in the 3-candidate pool (Tanimoto-0.90 regioisomer: 4-fluorophenyl / phenyl swapped). IR 1641, 1510 cm^{-1} ; ^{13}C 195.7. Verifier unused. This is a never-proposed miss.
Verify-save	v3-R25 / $\text{C}_{10}\text{H}_{14}\text{N}_2\text{O}$	True 2-pyridyl picolinamide at self-rank 3; self-rank 1 is 3-pyridyl. Forward-verify recovers (chamfer 0.42 vs 1.30 ppm). IR 1679 cm^{-1} . Figure S4. Not Listing 1.
Verify-fail	v3-R26 / $\text{C}_{14}\text{H}_{11}\text{N}_3\text{O}_3$	True 3-nitrophenyl dihydroquinazolinone in pool (self-rank 2); 2-nitro ranked first. Chamfer 1.35 vs 1.36 ppm. IR 1652, 1530, 1352 cm^{-1} .

9.1 Proposal miss — main-R06 / $\text{C}_{29}\text{H}_{30}\text{FNO}_2$

True acridinedione never enters the 3-candidate pool (Tanimoto-0.90 regioisomer: 4-fluorophenyl / phenyl swapped). IR 1641, 1510 cm^{-1} ; ^{13}C 195.7. Verifier unused. This is a never-proposed miss: re-ranking cannot recover a structure that was never proposed (Eq. 5).

9.2 Verify-save — v3-R25 / C₁₀H₁₄N₂O

True 2-pyridyl picolinamide at self-rank 3; self-rank 1 is 3-pyridyl. Forward-verify recovers (chamfer 0.42 vs 1.30 ppm, Eq. 6). IR 1679 cm⁻¹. Main-text Figure 5; SI Figure S4.

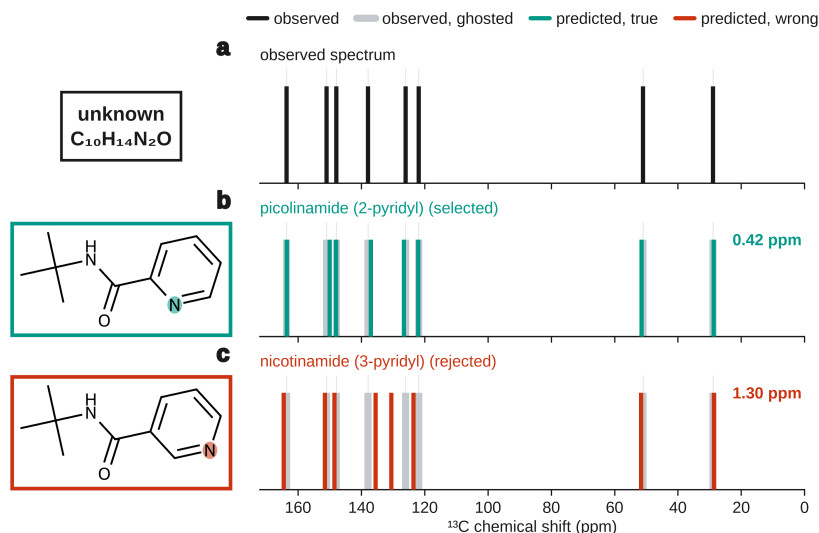


Figure S4: Locked v3-R25 verify-save (main-text Figure 5). Self-rank prefers 3-pyridyl; forward ¹³C recovers 2-pyridyl (chamfer 0.42 vs 1.30 ppm). A verify-save, not a recall win. Not Listing 1.

9.3 Verify-fail — v3-R26 / C₁₄H₁₁N₃O₃

True 3-nitrophenyl dihydroquinazolinone in pool (self-rank 2); 2-nitro ranked first. Chamfer 1.35 vs 1.36 ppm. IR 1652, 1530, 1352 cm⁻¹. Ranking can fail when predicted ¹³C lists are nearly tied. This is why we do not claim an accuracy advance from forward-verify (McNemar $p=0.55$; 7 gained, 4 lost on the protocol slice).

9.4 Listing 1 is the IRExp record plate

Main-text Listing 1 is listing1_irexp_record (C₁₂H₁₆N₂O₂ urea; InChIKey NTJIYHYWVZYBEX-UHFFFAOYSA-N). SMILES / InChIKey / licence fields are resource metadata. Bench solvers see only formula and printed peak lists. Gold constitution is scored only by Eq. 1. Locked v3-R25 remains the mechanism worked case (main-text Figure 5), not Listing 1.

The plate is reprinted as Figure S2.

9.5 Extra panels (already-reported numbers)

Formula-only collapse (control, $n=60$). Formula only: 3/60 (5%) top-1 and 3/60 (5%) recall. Formula + IR + ¹H + ¹³C: 14/60 (23%) / 19/60 (32%). Outcomes perfectly nested (McNemar $p=0.001$). The model is reading the peak lists, not only the formula.

Generate-wide (60-compound arm). Ten solver agents, up to six candidates each, pooled then re-ranked: recall 32% → **42%**, top-1 23% → **30%** (McNemar $p=0.34$); verification precision falls 84% → 72%. The training-free ceiling remains recall-limited. This is a success-for-recall panel, not an accuracy claim.

IRexp-fine-tuned generator (locked slice). Pooled with Claude: recall 33.5% $\rightarrow$ 54.1%, top-1 28.4% $\rightarrow$ 35.1% (McNemar $p=0.015$). The wall is a property of training-free elicitation, not of the task. Training is the one intervention already seen to move $P(\text{true} \in \text{pool})$.

Battery-electrolyte subset. $n=46$ literature electrolyte chemistry (not operando): 26% top-1, recall-bound — same regime as the locked slice.

Four-model Claude ladder ($n=24$). Haiku $\subset$ Sonnet $\subset$ Opus $\subset$ Fable is capability-sensitive but underpowered for adjacent ranks, with a disclosed protocol asymmetry. Not a headline.

Difficulty variation. Pre-registered draws vary in difficulty. That is a result. The headline remains $n=500$ (Eq. 7).

10 Prompt skeletons

10.1 Headline solver-facing skeleton

Wrapping instruction text was not captured on the headline Claude arm (Limitations (i); docs/-SUBMISSION.md gap #1). Only the per-compound four-field payload is released. The skeleton below is the contract, not a recovered verbatim system prompt.

You are given only:
formula (as from HRMS)
ir_bands_cm-1 (author-transcribed experimental list)
h_nmr, c_nmr (printed shift lists)
Return ≤ 3 ranked SMILES.
No name, no SMILES hint, no scaffold, no retrieved spectrum.
Check molecular formula with RDKit before emitting a candidate.

Listing 1 payload (fields only; ^1H abbreviated with ellipsis in the main float; gold held out):

```
{
  "formula": "C12H16N2O2",
  "ir_bands_cm-1": [3337.0, 2969.0, 1633.0, 1556.0,
                    1505.0, 1240.0, 1181.0, 1071.0, 826.0],
  "h_nmr": "8.07 (brs, 1H, NH), 7.25 (d, J = 7.3 Hz, 2H, Ar), ...",
  "c_nmr": "154.8, 154.4, 134.0, 119.9, 114.3, 55.5, 45.0, 31.8, 14.8"
}
```

10.2 Forward-verify skeleton

For each candidate SMILES, predict a ^{13}C peak list without seeing the observed list. Re-rank by $d(\hat{P}(c), P_{\text{obs}})$ (Eq. 6). The instruction dispatched to the forward-prediction sub-agents was likewise not captured; only the candidate SMILES and the resulting numeric lists are on disk for the protocol slice.

10.3 Cross-vendor harness

The verbatim prompts in the (unreleased-here) ESI S2 of the spectro-agent tree are the cross-vendor harness prompts, not the missing headline Claude wrapper. External submissions must document model version, prompt, and tool access (docs/LEADERBOARD.md).

10.4 Deposit schema

```
{ "qid": "R01", "candidates": ["SMILES_rank1", "SMILES_rank2", "SMILES_rank3"] }
```

Score locally (spectro-agent):

```
python scripts/score_submission.py --predictions my_run.jsonl --name "YourModel-1.0"
python scripts/score_pooled.py --stereo # validate-clean stereo extras
python scripts/forward_verify_all.py # protocol-slice diagnostic
```

11 Hyperparameters, hardware, and model cards

11.1 Headline model card

Table 10: Headline solver. Gaps are recorded, not filled.

item	value
headline solver	Claude Opus, consumer subscription
thinking tier (core protocol)	none; a labelled thinking-tier arm exists
snapshot / seed / temperature / top_p / top_k / max_tokens	not exposed; not recorded (SUBMISSION gaps #2-#3)
candidate budget	≤ 3 SMILES / qid
tool access	RDKit formula / parse check only; no web search
fine-tuning	none on the headline arm
paid API	no, for the core protocol
hardware	consumer harness; no cluster job for inference
scoring hardware	commodity CPU; RDKit InChIKey-14
cross-vendor arm	$n=60$; Grok 4.6, Gemini 3.7 Flash, GPT-5.6 Sol
Claude mean candidates on that arm	2.20 (others 3.00) — recall rankings approximate
expand-500 solver	Claude Opus thinking-tier (declared; not headline)
Fable incomplete arm	deposits not scored; not pooled
scoring scripts	scripts/score_pooled.py, score_main.py, benchmark_v2.py score2

11.2 Cross-vendor card ($n=60$)

Table 11: Cross-vendor decomposition. Recall and precision use different denominators.

model	generation recall	prec. recall	multi-candidate only
Claude Opus	19/60 = 32%	16/19 = 84%	10/13 = 77%
Grok 4.6	32/60 = 53%	20/32 = 62%	20/32 = 62%
Gemini 3.7 Flash	30/60 = 50%	22/30 = 73%	22/30 = 73%
GPT-5.6 Sol	25/60 = 42%	17/25 = 68%	16/24 = 67%

Verification precision exceeds generation recall in every arm. Bootstrap separates the paired gap for Claude (+52.5 points), GPT-5.6 Sol (+26.3) and Gemini (+23.3); Grok’s gap (+9.2) is directional. A clean-clone control for Grok bounds key leakage (McNemar $p=0.39$). Reasoning-effort tier per vendor arm was not recorded (gap #2).

11.3 Literature decomposition (copied, not re-derived)

Any system reporting top-1 = a and top- k = b implies recall $\geq b$ and conditional precision $\leq a/b$.
Script: scripts/literature_decomposition.py.

Table 12: Selected literature decomposition rows (as in main-text Table 6).

system (n)	top-1	top- k (k)	recall	prec.
this work, solver alone (500)	45.4%	49.8% (3)	49.8%	91.2%
this work + forward-verify (slice)	29.9%	33.5% (3)	33.5%	89.2%
NMR-Solver, literature (450)	52.9%	67.3% (10)	$\geq 67.3\%$	$\leq 78.6\%$
NMRAgent, literature (450)	61.6%	70.0% (10)	$\geq 70.0\%$	$\leq 88.0\%$
Espejo Morales, education (236)	80.9%	90.0% (5)	$\geq 90.0\%$	$\leq 89.9\%$
Espejo Morales, industrial (34)	20.6%	29.1% (5)	$\geq 29.1\%$	$\leq 70.9\%$
Alberts, 6–13 atoms, NIST IR (3,455)	63.8%	84.0% (10)	84.0%	75.9%
SpecX, scaffold split, simulated (99,439)	29.7%	50.6% (10)	50.6%	58.7%

Three groups changed only the data, and recall carried 68% (NMR-Solver simulated $\rightarrow$ real), 70% (SpecX random $\rightarrow$ scaffold) and 83% (Espejo education $\rightarrow$ industrial) of each collapse. The solver-alone precision column on $n=500$ is self-ranking (227/249); the forward-verify row is the instrumented protocol slice (58/65).

12 Two-paper split and anonymity

IRSpectra-Bench (this ICLR paper) is the blind peak-list benchmark and the propose $\gg$ verify diagnosis. IRExp is a companion Scientific Data Data Descriptor (in preparation): 121,233 records; 43,060 structure-linked; 33,201 full IR+ ^1H + ^{13}C +structure quadruples. This SI cites that resource and does not re-present Methods / Data Records / licence-pool narrative.

Double-blind review copy: <https://anonymous.4open.science/r/peaklist-corpus-review-10C4/>. Named Hugging Face / GitHub hosting is restored at camera-ready. Do not put ilkhamfy/* URLs in the PDF. Fine-tunes must use data/train_no_bench.jsonl.gz so benchmark InChIKeys are withheld.

Listing 1 is the only IRExp-style JSON+molecule plate in the ICLR PDF. It is a pointer, not a second paper.

13 Reproducibility checklist

ICLR-style checklist, filled honestly. "N/A" is used only when the item does not apply; gaps are named.

1. Code / data / frozen deposits. Frozen questions / predictions live in spectro-agent. This postcard repo is the ICLR manuscript. Camera-ready will name the archival host. Roster: frozen qid lists in the code release.
2. Headline regeneration. Committed score notes regenerate 227/500 and 249/500 (docs/HEADLINE_N500_2026-09-20.md).
3. $n=500$ fverify wall. spectro-agent data/fverify_n500/WALL_n500.md (PR #68) is 204 verified / 45 misranked / 251 never-proposed. Do not invent other integers.
4. Protocol-slice fverify. python scripts/forward_verify_all.py regenerates the protocol-slice diagnostic (not the $n=500$ wall).
5. expand-500 score2. Restore /tmp/blind/_key/benchmark_expand_500.answers2.jsonl.withheld, then benchmark_v2.py score2 -outdir data/benchmark_expand_500. Re-withhold the key.

Confirm git ls-files data/benchmark_expand_500/answers2.jsonl is empty.

6. Do not commit withheld answers2.jsonl keys.
7. Do not invent CIs for expand-500; -ci was not run.
8. expand-500 fverify score is already in NIGHT_POOL / fverify STATUS (103/230 verify vs 129/230 self). Do not invent CIs or a second wall. Do not re-run official prep.
9. Double-blind. Companion bib is Anonymous; dataset URL in the PDF is only the anonymous.4open review copy.
10. Fine-tunes must use data/train_no_bench.jsonl.gz.
11. IRexp construction, licences, and transcription audit belong in the Scientific Data companion — not in this SI beyond Listing 1.
12. Inference is not bit-exact. No pinned snapshot, temperature, or seed (Limitations (i)). Distributional agreement of frozen deposits is the intended claim.
13. Uncaptured artefacts (cannot be recovered): headline Claude wrapper text; dated model snapshots; decoding parameters; main-round sampler seeds; closed-book audit transcripts (docs/SUBMISSION.md gaps #1–#5).
14. Deferred arms. Expert-chemist audit not run. Leave-one-modality-out abandoned. Parser extraction-recall audit deferred to Sci. Data. SpectraLLM / IR-Agent / Spectro / NMIRacle / Alberts / CASE not scored on this bench.
15. Figure rebuild limits. fig_wall_diagnostic reprints the $n=500$ forward-verify wall (204/45/251). Fig. 1 is the $n=500$ fverify plate (249/500 recall $\times$ 204/249 verified | pool = 40.8% verified; wall 204/45/251). Do not rebuild the wall from a single-arm diagnosis.json.
16. This manuscript compile. python3 scripts/build_pdf.py (main); python3 scripts/build_si.py (this SI). Engine: tectonic 0.17.0 or pdfLaTeX+BibTeX / XeLaTeX+BibTeX.

13.1 What a reviewer can re-check without the withheld keys

- Every a/b ($c\%$) in the main text and this SI is internally consistent with the integers in docs/HEADLINE_N500_2026-09-20.md, docs/POOLED_HEADLINE_2026-09-16.md and docs/NIGHT_POOL_2026-09-20.md and spectro-agent data/fverify_n500/WALL_n500.md.
- Fig. 1 caption, Limitations (vi), and Appendix FAQ agree: Fig. 1 = $n=500$ fverify (249/500 recall $\times$ 204/249 verified | pool = 40.8% verified; wall 204/45/251); generation headline remains 227/500 top-1 and 249/500 recall.
- expand-500 all-230 rows carry “no CIs / thinking-tier / not a second wall”. Official arm fverify is 103/230 verify vs 129/230 self (SI only). Do not quote 243/519.

14 Reviewer FAQ (SI copy)

These answers restate main-text Limitations and Appendix B. They add no new metrics.

Why is top-1 45% if the wall figure looks worse? Generation remains 227/500 (45.4%) top-1 and 249/500 (49.8%) recall (self-rank 227/249). Figure 1 is the $n=500$ fverify plate: 249/500 recall $\times$ 204/249 verified | pool = 40.8% verified; wall 204/45/251. The appendix wall reprints that plate (data/fverify_n500/WALL_n500.md).

Is the roster a hidden mix? The $n=500$ roster is the frozen qid list in the code release. Do not reconstruct a composition story from protocol notes.

Where is the instrumented forward-verify wall? The $n=500$ wall is 204/45/251 (§8). The older protocol-slice diagnostic is also here. Do not invent CIs.

Where is expand-500 official fverify? This SI: 103/230 verify vs 129/230 self (§8); no CIs; not a second wall. Headline generation is $n=500$ (Eq. 7).

Consumer harness / no snapshot / no thinking tier? Correct and intentional for the scored headline. Scoring of frozen deposits is reproducible; inference is not bit-exact.

Why no SpectraLLM / IR-Agent / CASE SOTA table? Different objects. None is scored on this bench. The paper is a stage diagnosis, not a method race.

Is IRexp a second ICLR paper? No. IRexp is a companion resource pointer (Listing 1; §12).

Should the headline move off $n=500$ or quote 243/519? No. Eq. 7 uses $\mathcal{S}$ of 500. Do not quote 243/519.

What this SI is not

A Data Descriptor. An expand-500 wall. A generation-composition story. A recommendation to switch the ICLR headline off $n=500$ or to quote 243/519. A source of invented CIs or new metrics. The $n=500$ forward-verify wall (204/45/251) is cited from data/fverify_n500/WALL_n500.md.